

SYSTEM INSTALLATION & CONFIGURATION GUIDE

Virtual Machine Deployment and Boot Architecture Analysis

Prepared By: Mar Aldrei Y. Avila

Course: System Administration and Maintenance

Section: BSIT 4A

Date: August 15, 2026

1. PROJECT OVERVIEW

This technical installation guide documents the step-by-step procedure for deploying and configuring a virtualized Operating System environment. The lab emphasizes hypervisor setup, baseline OS configuration, and command-line verification. Additionally, this guide provides a comparative breakdown between Legacy BIOS and UEFI firmware architectures, mapping out the low-level execution sequence of system boot processes.

2. OBJECTIVES

- Deploy a functional Linux/Windows virtual environment using standard virtualization software.
- Configure essential virtual machine resources including CPU, RAM, virtual storage, and network interfaces.
- Perform post-installation system verification using CLI diagnostic tools.
- Analyze low-level system boot flows comparing Legacy BIOS (MBR) and modern UEFI (GPT/ESP) standards.

3. SOFTWARE REQUIREMENTS

- **Host Operating System:** Windows 10/11 64-bit Architecture
- **Hypervisor:** Oracle VM VirtualBox Version 7.0+ / VMware Workstation
- **Target ISO Images:** Ubuntu Server 22.04 LTS / Windows Server 2022
- **Diagramming Software:** Draw.io / Web Browser

4. VIRTUAL MACHINE SPECIFICATIONS

Hardware Parameter	Allocated Value
Base Memory (RAM)	4096 MB (4 GB)
Processors (vCPU)	2 Execution Cores
Virtual Disk Size	25.00 GB (Dynamically Allocated VDI)
Storage Controller	SATA Controller / AHCI Mode
Network Adapter	NAT / Bridged Adapter

5. STEP-BY-STEP INSTALLATION PROCEDURE

1. **Virtual Machine Creation:** Open VirtualBox, click *New*, assign the VM name, choose Linux/Windows type, and select the target 64-bit version.
2. **Resource Allocation:** Slide RAM allocation to 4096 MB and assign 2 vCPU cores.
3. **Virtual Hard Disk Creation:** Select "Create a Virtual Hard Disk Now", choose VDI format, set it to Dynamically Allocated, and set the disk size to 25 GB.
4. **Mounting Installation Media:** Navigate to *Settings > Storage*, select the empty Optical Drive, and browse to choose the downloaded OS ISO file.
5. **OS Execution & Setup:** Power on the VM, follow on-screen prompt selectors for language, keyboard layout, network interface setup, and disk partitioning.
6. **Account Creation:** Specify administrator username, password, hostname, and system credentials. Complete installation and reboot the VM.

Figure 1: Virtual Machine Initial OS Installation Process.

6. CONFIGURATION DETAILS

- **Package Repository Updates:** Updated system package index to latest security patches using package manager commands.
- **Hypervisor Integration:** Installed Guest Additions / Open VM Tools to enable dynamic display resizing and bidirectional clipboard integration.
- **Network Interface Configuration:** Set up DHCP/Static IP configuration and verified DNS resolution.

7. VERIFICATION COMMANDS

System operational status was validated using the following terminal commands:

```
# Verify Active Linux Kernel Version
uname -r

# Verify Storage Partitions and Mount Points
lsblk

# Check Memory Resource Utilization
free -h

# Verify Firmware Boot Mode (UEFI vs BIOS)
[ -d /sys/firmware/efi ] && echo "UEFI Mode" || echo "Legacy BIOS Mode"
```

Figure 2: Terminal Output Confirming System Specs and Active Services.

8. BIOS VS UEFI COMPARISON

Feature / Characteristic	Legacy BIOS	Modern UEFI
Processor Mode	16-bit Real Mode	32-bit or 64-bit Native Mode
Partition Scheme	Master Boot Record (MBR)	GUID Partition Table (GPT)
Maximum Partition Capacity	2 Terabytes (TB)	9.4 Zettabytes (ZB)
Primary Partitions	Maximum of 4	Up to 128 Partitions
Boot Sector Location	Sector 0 (MBR)	EFI System Partition (ESP)
Security Features	None	Secure Boot Integration

9. BOOT PROCESS FLOWCHART

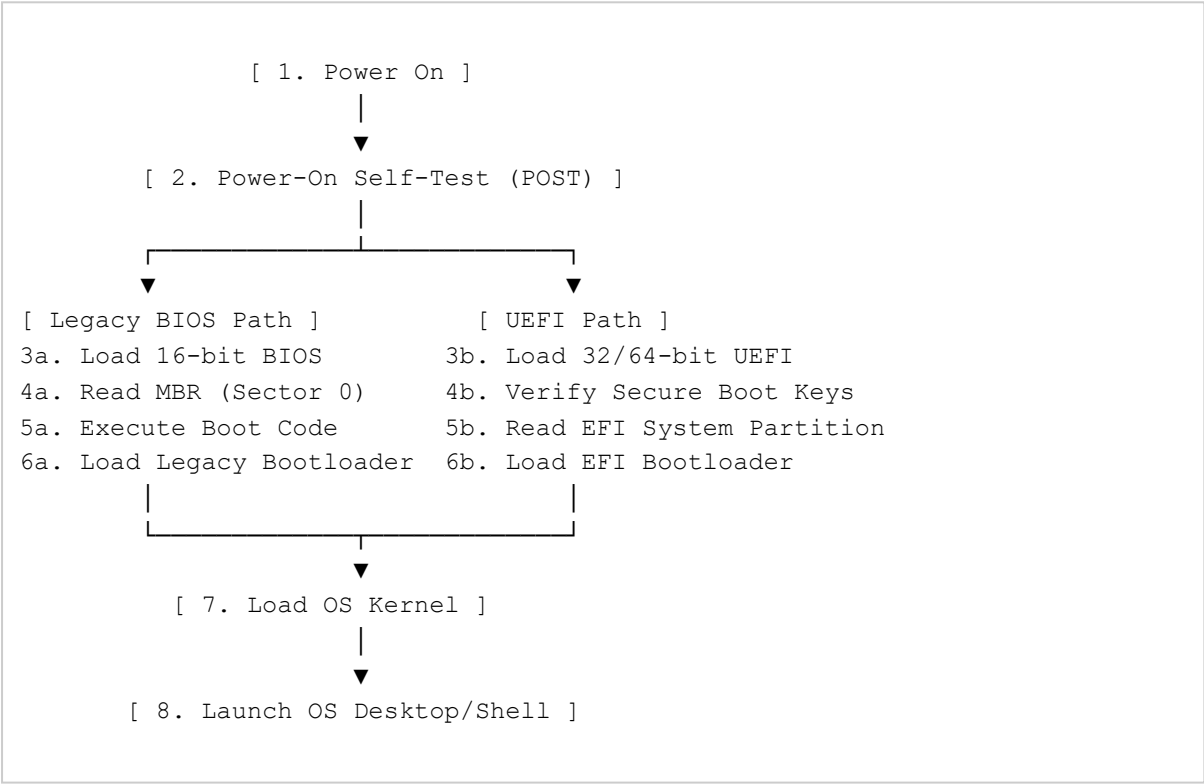


Figure 3: System Boot Process Execution Diagram.

10. WINDOWS SERVER INSTALLATION SUMMARY

In addition to Linux deployment, a Windows Server 2022 instance was initialized. The setup involved selecting the Server Core / Desktop Experience image, configuring custom disk volumes via Setup Wizard, defining the local Administrator password, and verifying Server Manager dashboard initialization.

11. OS COMPARISON TABLE

Metric / Feature	Ubuntu Server	Windows Server
Default Management Interface	Command-Line Interface (CLI)	Graphical User Interface (GUI / Server Manager)
Base Resource Consumption	Low (~512MB RAM minimal)	Moderate to High (~2GB RAM baseline)
License Model	Open Source (Free / GNU GPL)	Proprietary Commercial Licensing
Package / Role Management	APT Package Manager	Server Manager / PowerShell Modules

12. PROBLEMS ENCOUNTERED

- **Hardware Virtualization Error:** Host system threw a VT-x/AMD-V virtualization error preventing VM initialization.
- **Display Resolution Constraints:** Guest OS was locked to low screen resolutions (800x600).

13. SOLUTIONS IMPLEMENTED

- Rebooted host machine, entered Host BIOS settings, and enabled Intel Virtualization Technology (VT-x) / AMD-V.
- Built and installed Hypervisor Guest Additions drivers to enable dynamic guest scaling and seamless cursor integration.

14. LESSONS LEARNED

This deployment project reinforced the necessity of structured virtual sandboxes for IT operations. Understanding low-level boot workflows—from POST execution to kernel loading—provides essential troubleshooting foundation for system administrators managing enterprise servers, firmware integrity, and partitioned storage systems.

15. REFERENCES

Arch Linux Wiki. (2024). *Arch boot process*. Arch Linux Documentation.
https://wiki.archlinux.org/title/Arch_boot_process

Microsoft. (2023). *Overview of boot options in Windows*. Microsoft Learn.
<https://learn.microsoft.com/en-us/windows-hardware/drivers/devtest/boot-options-in-windows>

Silberschatz, A., Galvin, P. B., & Gagne, G. (2018). *Operating system concepts* (10th ed.). John Wiley & Sons.